

Understanding Analysis, 2nd Edition

Chapter 7 - Section 7.2 - The Riemann Integral

1. Let f be a bounded function on $[a, b]$, and let P be an arbitrary partition of $[a, b]$. First, explain why $U(f) \geq L(f, P)$. Now, prove Lemma 7.2.6: for any bounded function f on $[a, b]$, it is always the case that $U(f) \geq L(f)$.

Solution: We'll prove it by contradiction. Suppose that $U(f) < L(f, P)$. By definition of infimum, there must exist a partition P_2 such that $U(f, P_2) < L(f, P)$, for otherwise the infimum would've been at least $L(f, P)$.

But $U(f, P_2) < L(f, P)$ contradicts Lemma 7.2.4: $L(f, P_1) \leq U(f, P_2)$ for any partitions P_1, P_2 .

To use this fact to prove Lemma 7.2.6, note that $L(f)$ is the supremum of all lower sums while $U(f)$ is an upper bound. Since by definition, an upper bound is *at least* the supremum, it follows that $U(f) \geq L(f)$.

Alternatively, we can prove Lemma 7.2.6 directly by contradiction as well. Suppose $U(f) < L(f)$ and let m be the midpoint i.e. $m = \frac{L(f) + U(f)}{2}$. By definition of infimum, there exists a partition P_1 such that $U(f, P_1) < m$. By definition of supremum, there exists a partition P_2 such that $m < L(f, P_2)$. But $U(f, P_1) < m < L(f, P_2)$ contradicts Lemma 7.2.4.

2. Consider $f(x) = 1/x$ over the interval $[1, 4]$. Let P be the partition consisting of the points $\{1, 3/2, 2, 4\}$.
 - (a) Compute $L(f, P)$, $U(f, P)$ and $U(f, P) - L(f, P)$.
 - (b) What happens to the value of $U(f, P) - L(f, P)$ when we add the point 3 to the partition?
 - (c) Find a partition P' of $[1, 4]$ for which $U(f, P') - L(f, P') < 2/5$.

Solution:

$$(a) \quad L(f, P) = (2/3 * 1/2) + (1/2 * 1/2) + (1/4 * 2) = 13/12.$$

$$U(f, P) = (1 * 1/2) + (2/3 * 1/2) + (1/2 * 2) = 11/6.$$

$$U(f, P) - L(f, P) = 3/4.$$

- (b) It gets smaller, because adding a new point to a partition decreases the upper sum while increasing the lower sum (Lemma 7.2.3). The new values are:

$$L(f, P) = (2/3 * 1/2) + (1/2 * 1/2) + (1/3 * 1) + (1/4 * 1) = 14/12.$$

$$U(f, P) = (1 * 1/2) + (2/3 * 1/2) + (1/2 * 1) + (1/3 * 1) = 5/3.$$

$$U(f, P) - L(f, P) = 1/2.$$

- (c) I initially solved this by adding more points to the initial partition + some random guessing, but chatgpt suggested a smarter way.

Let's create an equally-spaced partition. Then we have

$$U - L = \Delta x \sum_{i=1}^n M_i - m_i$$

For $f(x) = 1/x$, the supremum $M_i = f(x_{i-1})$ and the infimum $m_i = f(x_i)$.

The sum telescopes:

$$\begin{aligned} U - L &= \Delta x [f(x_0) - f(x_1) + f(x_1) - f(x_2) + \dots + f(x_{n-1}) - f(x_n)] \\ &= \Delta x (f(x_0) - f(x_n)) \end{aligned}$$

We have $x_0 = 1$ and $x_n = 4$, resulting in $\Delta x(1 - 4) = \frac{3}{4}\Delta x$ and we want this to be less than $\frac{2}{5}$:

$$\frac{3}{4}\Delta x < \frac{2}{5} \implies \Delta x < \frac{8}{15}$$

We can pick $\Delta x = \frac{1}{2}$. So the final partition P' is $\{1, \frac{3}{2}, 2, \frac{5}{2}, 3, \frac{7}{2}, 4\}$.